Transmission of Squeezed Light in Atmospheric Channels

Specialization Elective & Core Project

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April 29, 2026

Motivation

- Need for **ultra-secure communication** in the quantum era
- **Quantum communication** enables fundamentally secure data transfer
- **Free-space channels** are essential for global quantum networks
- However, the **atmosphere is unstable**
- **Atmospheric turbulence** causes:
  - Beam wandering
  - Intensity fluctuations
  - Phase distortions
- These effects **degrade fragile quantum states** (e.g., squeezed light)
- **Key Question:**
  - How do we model these effects?
  - What happens in real transmission?

Modeling Atmospheric Effects

- Atmospheric turbulence is **random and time-varying**
- Exact prediction is not possible
  $\Rightarrow$ Use **statistical models**
- These models describe:
  - Fluctuations in transmission
  - Beam distortion effects
- Different models apply to different regimes:
  - Weak turbulence
  - Strong turbulence
- **Key Question:** Which model best describes real atmospheric channels?

Squeezed Light: Basic Concept

Definition:

- Squeezed light is a non-classical state of light where quantum noise is reduced in one quadrature
- This reduction comes at the cost of increased noise in the conjugate quadrature

Key Idea:

- Heisenberg Uncertainty Principle governs the trade-off:

$$\Delta X \cdot \Delta P \geq \frac{\hbar}{2}$$

- Squeezing reduces uncertainty in one variable below the standard quantum limit

Importance:

- Enables precision measurements beyond classical limits
- Used in interferometry and quantum communication

Quadratures and Noise Reduction

Field Quadratures:

- The electromagnetic field can be described using two quadratures:
- In-phase component (X) and out-of-phase component (P)

Quantum Uncertainty:

$$\Delta X \cdot \Delta P \geq \frac{\hbar}{2}$$

- In classical light: noise is equally distributed
- In squeezed light: noise is reduced in one quadrature

Noise Reduction:

- One quadrature has reduced uncertainty (squeezed)
- The other has increased uncertainty (antisqueezed)
- Total uncertainty still satisfies quantum limit

Need for Statistical Models

Atmospheric Nature:

- Atmospheric turbulence is random and time-varying
- Causes fluctuations in intensity and phase of light

Why Not Deterministic Models?

- Exact prediction of atmospheric behavior is not possible
- Environmental factors change continuously (temperature, pressure, wind)

Role of Statistical Models:

- Describe fluctuations using probability distributions
- Model transmission variations and noise effects
- Help analyze degradation of squeezed states

Key Idea:

- Different models are used for different turbulence regimes

Statistical Models Used in This Study

Context:

- Atmospheric channel characterized by fluctuating transmittance
- Described using Probability Distribution of Transmittance (PDT)

Models Considered:

- **Log-Normal Model**
  - Models weak turbulence
  - Based on small intensity fluctuations
- **Beam Wandering Model**
  - Accounts for random beam displacement
  - Important in free-space channels
- **Elliptic Beam Model**
  - Includes beam distortion and shape changes
- **Combined Models (Law of Total Probability)**
  - Combines beam wandering + intensity fluctuations
- **Beta Distribution Model**
  - Empirical model fitting wide range of conditions

Objective of the Study

Goal:

- To analyze how atmospheric turbulence affects transmission of squeezed light

Approach:

- Use numerical simulations of atmospheric channels
- Compare different statistical models with simulated data

Key Question:

- Which model best describes real atmospheric transmission?

Evaluation Metrics

1. **Probability Distribution of Transmittance (PDT):**
   - Describes how transmission fluctuates due to atmospheric turbulence
   - Key characteristic of atmospheric quantum channels

2. **Kolmogorov-Smirnov (KS) Statistic:**
   - Measures difference between model prediction and simulated data
   - Lower KS value $\rightarrow$ better model accuracy

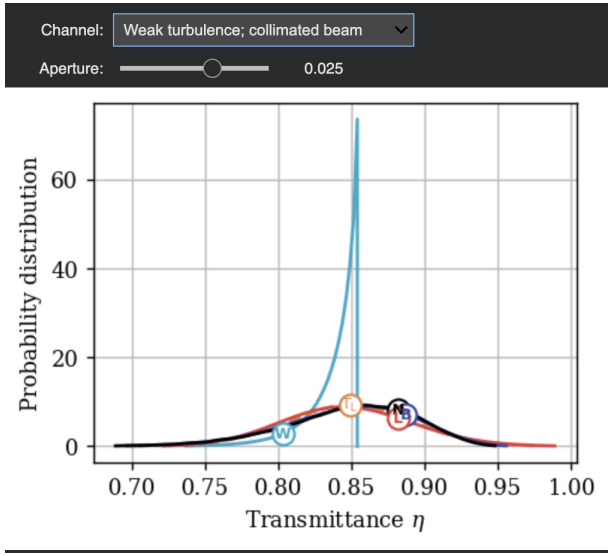
Purpose:

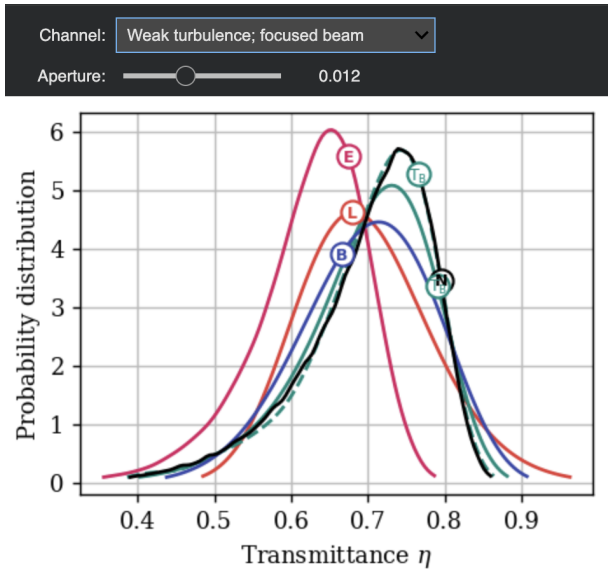
- Compare different models based on how well they match real channel behavior

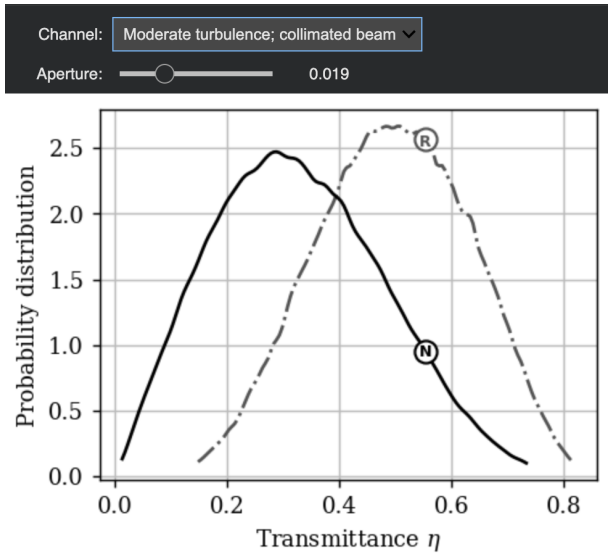
PDT under Weak and Moderate Turbulence

Observation:

- Transmission fluctuations vary with turbulence strength
- Beam type (coherent vs focused) affects distribution shape







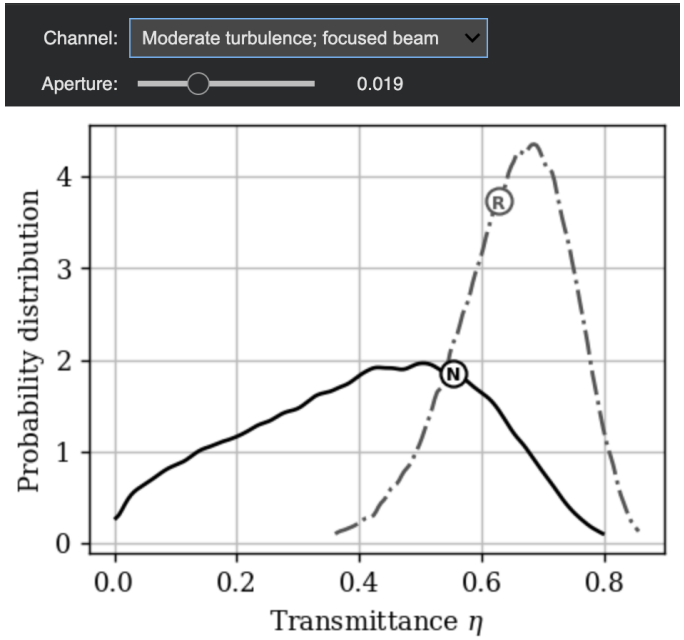
Insight:

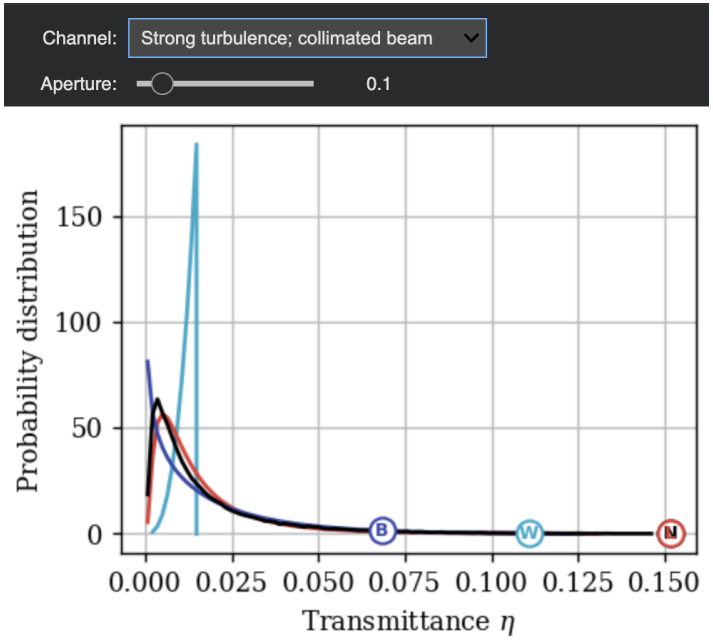
- Weak turbulence $\rightarrow$ narrow distributions
- Moderate turbulence $\rightarrow$ broader fluctuations

PDT under Strong Turbulence Conditions

Observation:

- Strong turbulence causes large transmission fluctuations





Insight:

- Distributions become highly skewed

Insights from PDT Analysis

Weak Turbulence:

- Log-normal and Beta models closely match simulated data
- Total probability model also performs well

Moderate Turbulence:

- Significant deviation between models and simulated data
- Distributions become broader with no sharp peak

Strong Turbulence:

- Transmittance shifts to very low values
- Beta and log-normal give closer approximation
- Beam wandering model performs poorly

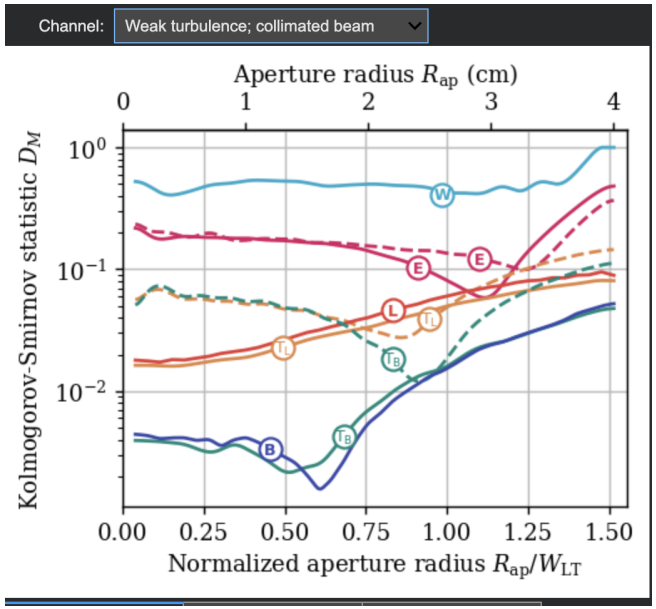
Overall Observation:

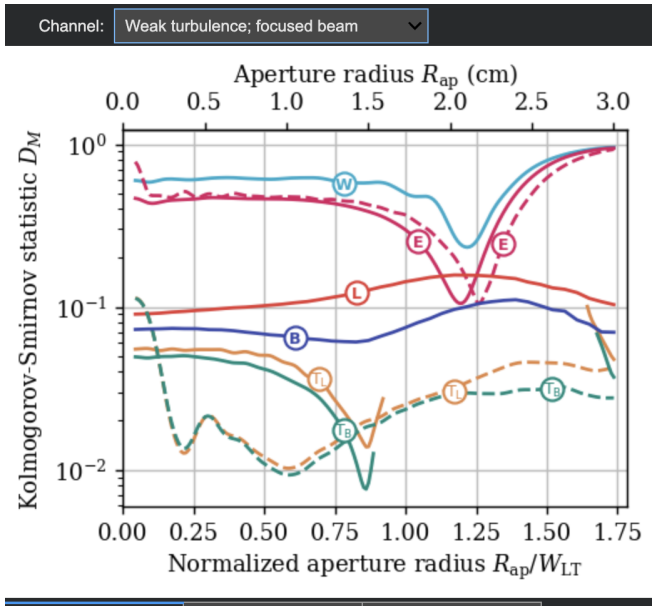
- Model accuracy strongly depends on turbulence conditions
- No single model works for all regimes

KS Statistic: Weak Turbulence

Observation:

- Lower KS value indicates better agreement with simulation





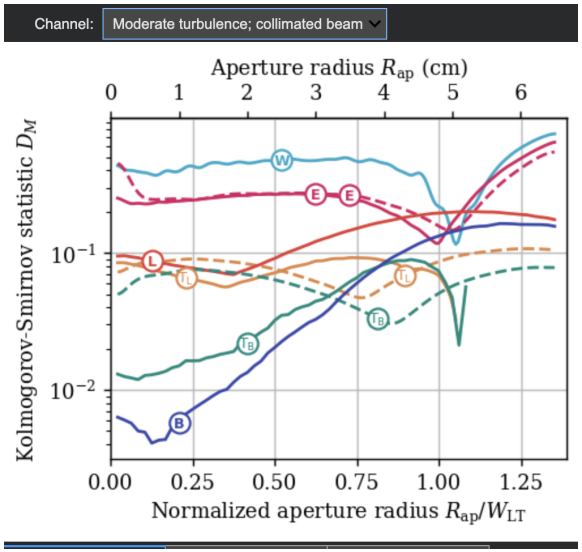
Insight:

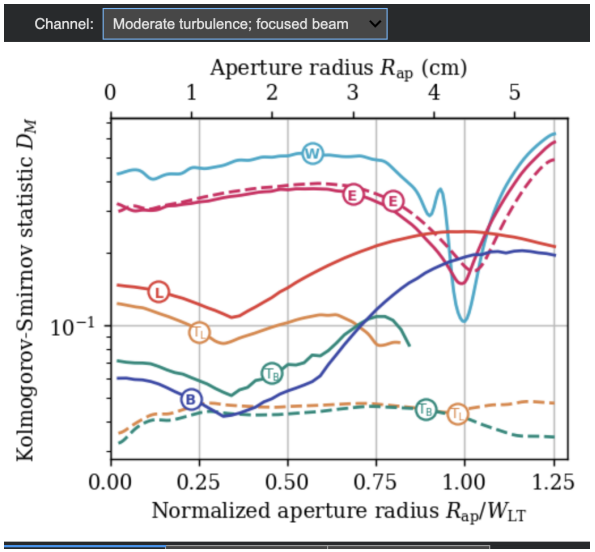
- Beta and total probability models perform best
- Log-normal works reasonably well in weak turbulence

KS Statistic: Moderate Turbulence

Observation:

- KS statistic varies with aperture and beam type
- Lower values indicate better agreement with simulation

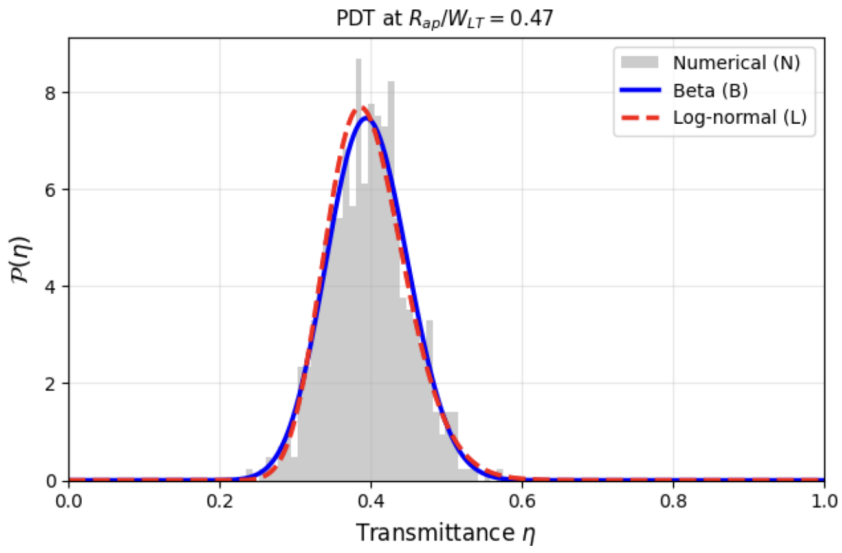


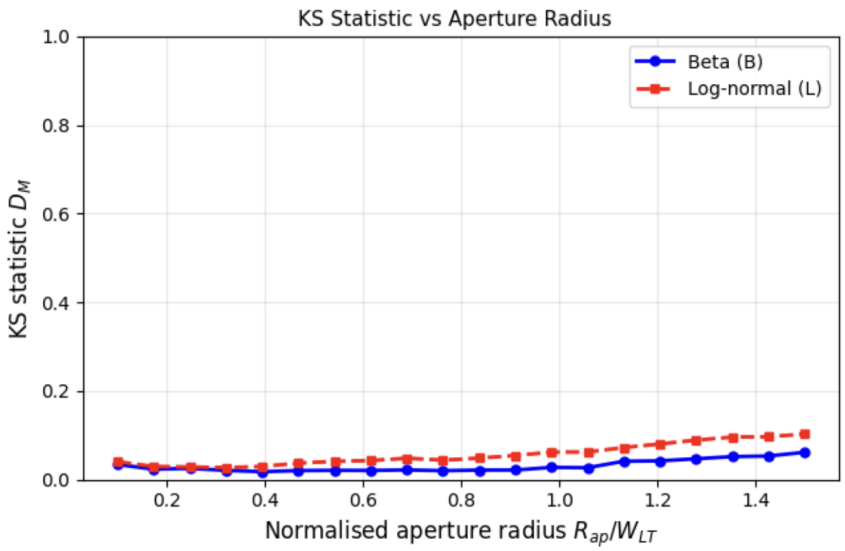


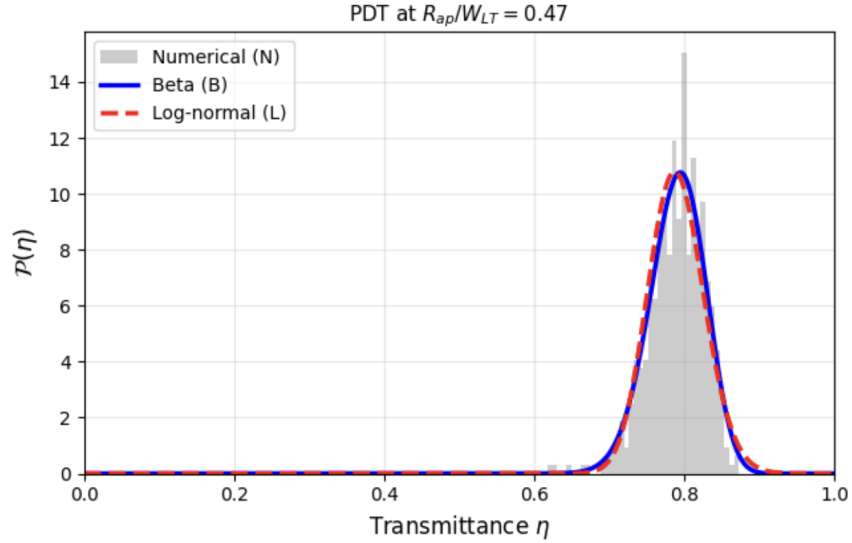
Insight:

- Beta distribution consistently provides the best fit
- Total probability models also perform well across conditions
- Log-normal and simpler models fail in moderate turbulence
- Model accuracy depends on aperture size (optimal region exists)
- Focused beams show larger deviations due to stronger distortions

Plots


PDT at $R_{ap}/W_{LT} = 0.47$
Numerical (N)
Beta (B)
Log-normal (L)
$\mathcal{P}(\eta)$
Transmittance η


KS Statistic vs Aperture Radius
Beta (B)
Log-normal (L)
KS statistic D_M
Normalised aperture radius R_{ap}/W_{LT}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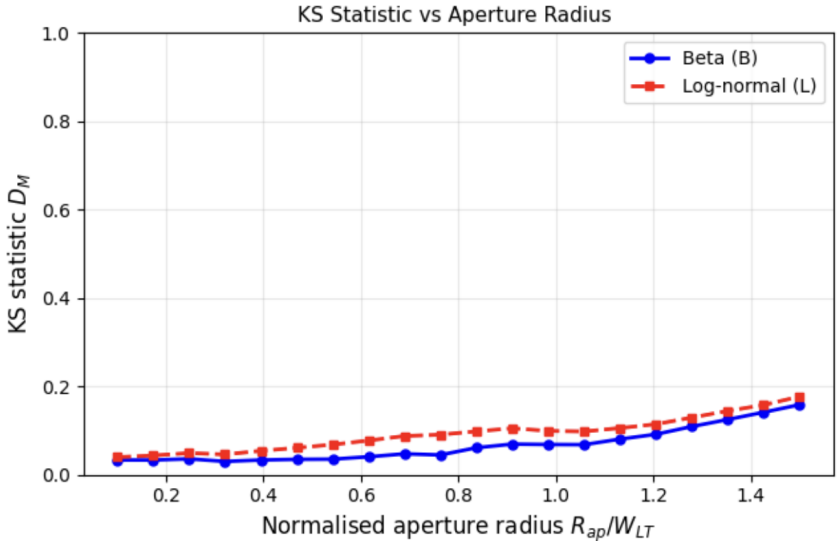
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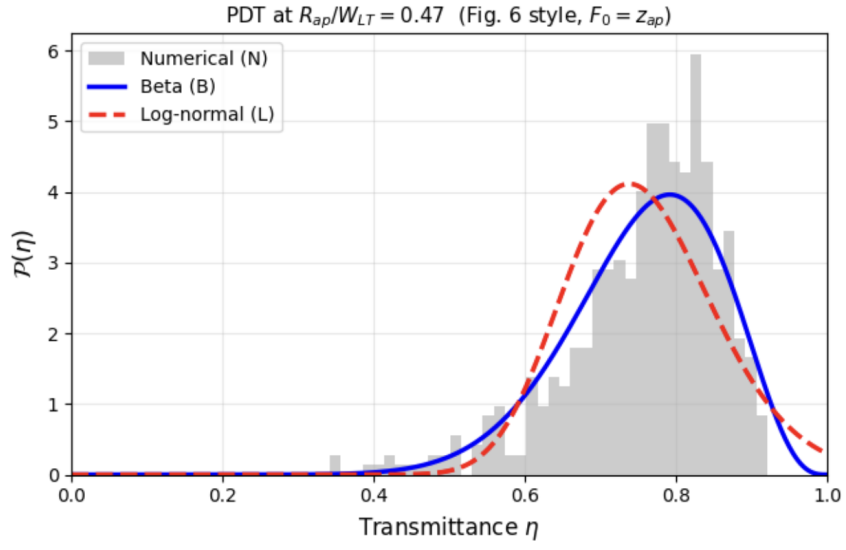
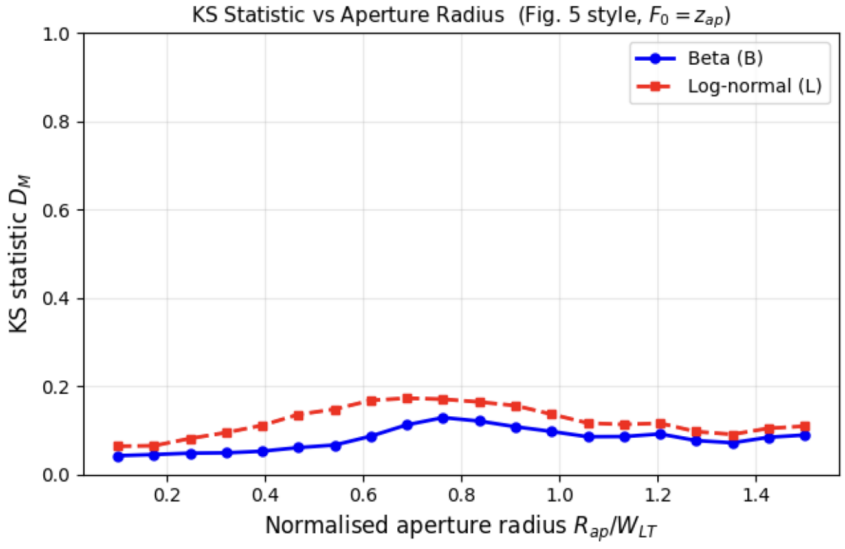
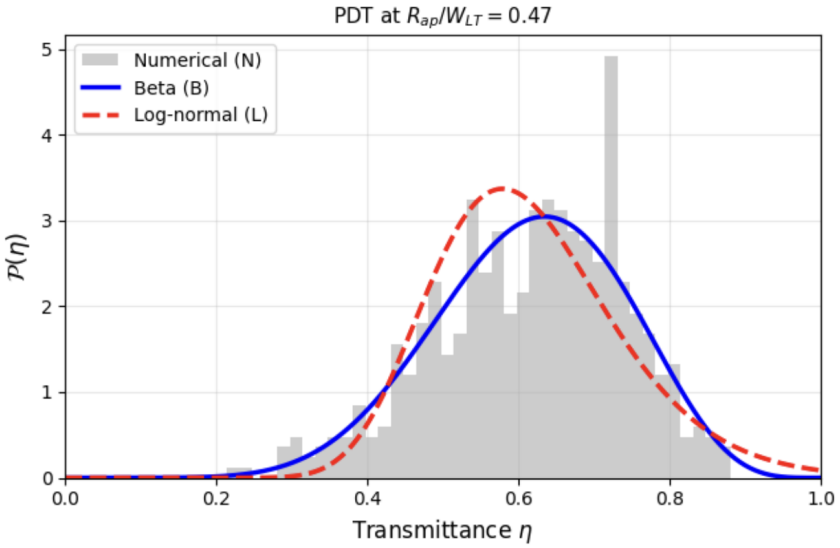
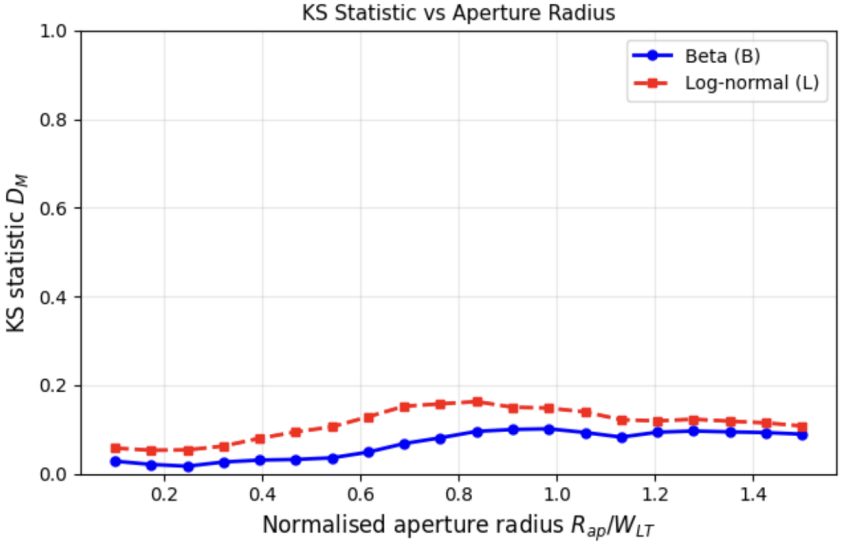
Normalised aperture radius R_{ap}/W_{LT}

Plots: Moderate Turbulence









Experimental Setup: Free-Space Quantum Channel

Overview:

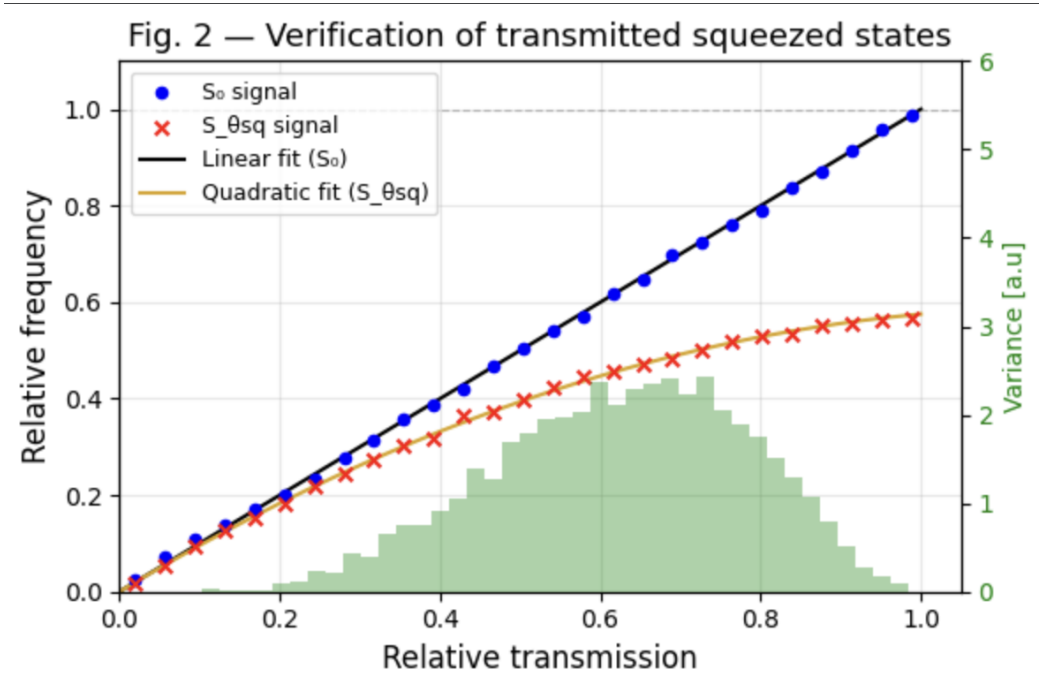
- Free-space atmospheric channel of length ~ 1.6 km (urban environment)
- Sender (Alice): Generates polarization-squeezed states using Kerr nonlinearity in optical fiber
- Transmission affected by atmospheric turbulence (beam wandering, distortion, loss)
- Receiver (Bob): Measures Stokes parameters via polarization detection

Key Features:

- Polarization encoding used to mitigate phase fluctuations
- Homodyne-like detection via Stokes observables
- Transmission fluctuations recorded as classical side information
- Postselection protocol applied to enhance observed squeezing

The experiment demonstrates the distribution of continuous-variable squeezed states through a turbulent atmospheric channel.

Verification of Squeezed State Transmission

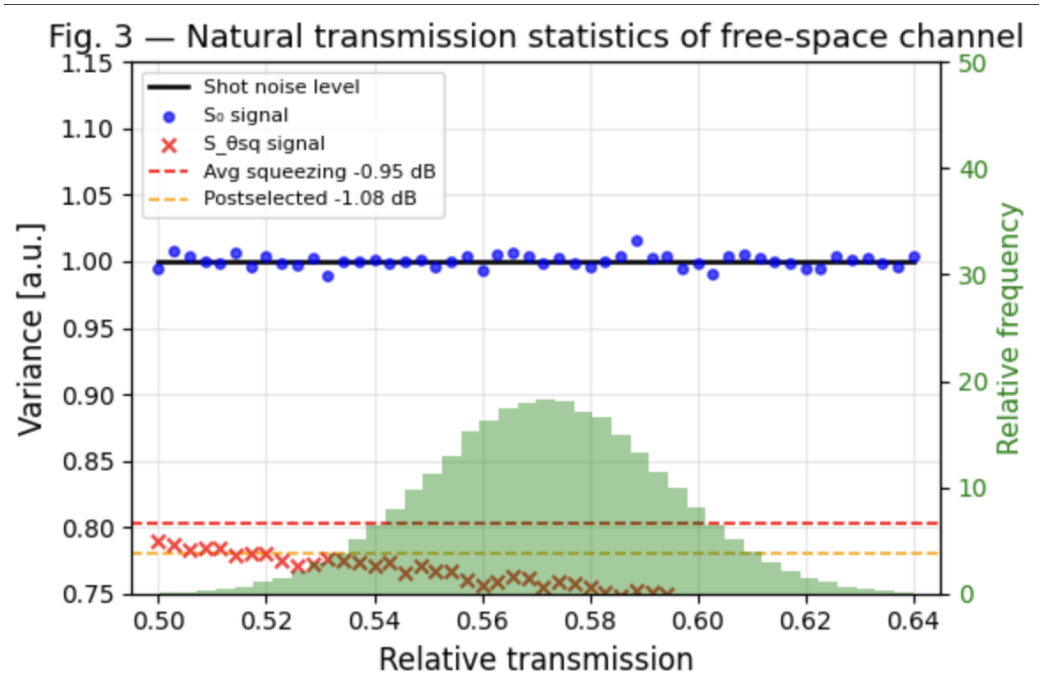


Key Observations:

- S_0 (shot noise) scales linearly
- $S_{\theta_{sq}}$ shows nonlinear behavior
- Signal remains below shot noise
- Confirms preservation of squeezing

Squeezed states survive atmospheric transmission but degrade with loss.

Transmission Statistics of Atmospheric Channel



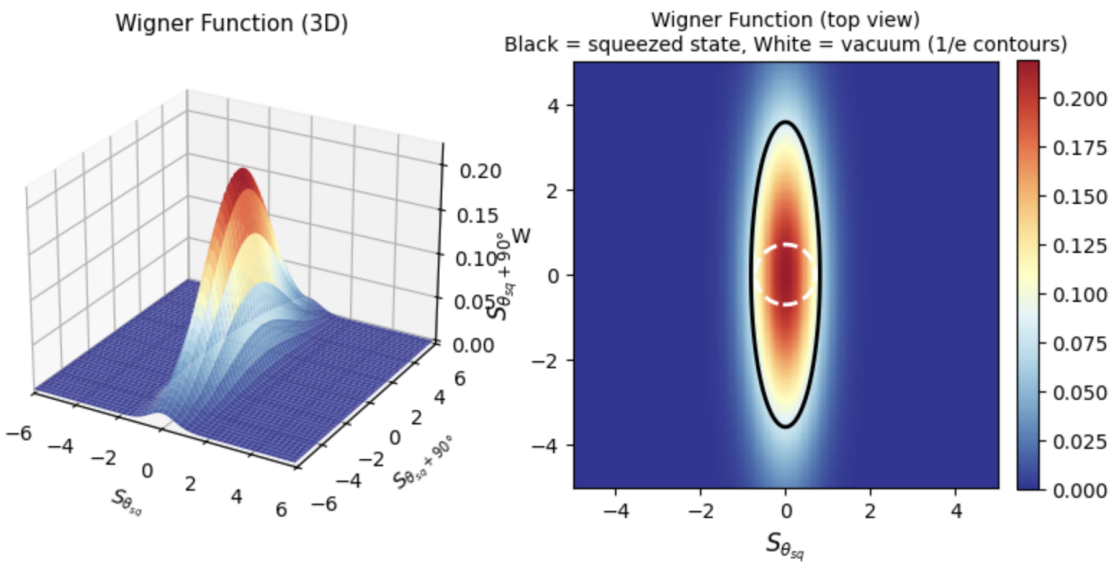
Key Observations:

- Transmission follows a statistical distribution (fading channel)
- Caused by turbulence: beam wandering, distortion, spreading
- Squeezing degrades at lower transmission
- Postselection improves squeezing performance

Atmospheric fluctuations lead to random transmission variations, requiring statistical modeling of quantum channels.

Reconstructed Wigner Function (After Atmospheric Channel)

Fig. 4 — Reconstructed Wigner function after 1.6 km atmospheric channel



Key Observations:

- Wigner function represents quantum state in phase space
- Elliptical shape → signature of squeezing
- Black contour: squeezed state
- White contour: vacuum reference
- Elongation shows noise redistribution between quadratures

Even after atmospheric transmission,

Linking Theory and Experiment

Key Connections:

- Atmospheric channel introduces random transmission fluctuations
- Statistical models (PDD, KS) capture these fluctuations (Part A)
- Experimental results confirm fading channel behavior (Fig. 3)
- Squeezing degrades nonlinearly with transmission (Fig. 2)
- Wigner function shows preservation of quantum state structure (Fig. 4)

Key Insight:

- Atmospheric quantum channels must be modeled statistically
- Combination of modeling and experiment provides a complete understanding

This unified approach validates both the theoretical models and experimental observations of squeezed light transmission.

Conclusion

Summary of Work:

- Studied transmission of squeezed light through atmospheric channels
- Analyzed statistical models (PDD, KS) for turbulence effects
- Replicated experimental results from PRL study
- Verified survival of squeezing under atmospheric propagation

Key Findings:

- Atmospheric channels behave as fading (statistical) channels
- Squeezing degrades with transmission loss but remains non-classical
- Postselection enhances observed squeezing
- Wigner function confirms preservation of quantum state structure

This work demonstrates the feasibility of continuous-variable quantum communication through atmospheric channels.